Circular Measure

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Converting between degrees and radians**

Convert 135° to radians, in terms of π.

$135^\circ \times \frac{\pi}{180^\circ} = \frac{3\pi}{4}$.

2 **[Foundational]** • **Target: Converting between degrees and radians**

Convert $\frac{5\pi}{6}$ radians to degrees.

$\frac{5\pi}{6} \times \frac{180^\circ}{\pi} = 150^\circ$.

3 **[Foundational]** • **Target: Converting between degrees and radians**

Convert 210° to radians, in terms of π.

$210^\circ \times \frac{\pi}{180^\circ} = \frac{7\pi}{6}$.

4 **[Foundational]** • **Target: Converting between degrees and radians**

Convert $\frac{7\pi}{4}$ radians to degrees.

$\frac{7\pi}{4} \times \frac{180^\circ}{\pi} = 315^\circ$.

5 **[Foundational]** • **Target: Arc length**

A circle has radius 8 cm. Find the length of an arc subtending an angle of $\frac{\pi}{3}$ radians at the center, using $s = r\theta$.

$s = r\theta = 8 \times \frac{\pi}{3} = \frac{8\pi}{3}$ cm.

6 **[Foundational]** • **Target: Arc length**

A circle has radius 12 cm. Find the length of an arc subtending an angle of $\frac{5\pi}{6}$ radians at the center, using $s = r\theta$.

$s = r\theta = 12 \times \frac{5\pi}{6} = 10\pi$ cm.

7 **[Foundational]** • **Target: Sector area**

Find the area of a sector with radius 6 cm and angle $\frac{2\pi}{3}$ radians, using $A = \frac{1}{2}r^2\theta$.

$A = \frac{1}{2}(6)^2\left(\frac{2\pi}{3}\right) = \frac{1}{2}(36)\left(\frac{2\pi}{3}\right) = 12\pi$ cm^2.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Sector area**

A sector has radius 10 cm and area 40 cm^2. Find the angle θ in radians.

$40 = \frac{1}{2}(10)^2\theta = 50\theta$, so $\theta = \frac{40}{50} = 0.8$ radians.

9 **[Intermediate]** • **Target: Sector area**

Find the area of a sector with radius 9 cm and angle $\frac{\pi}{6}$ radians, using $A = \frac{1}{2}r^2\theta$.

$A = \frac{1}{2}(9)^2\left(\frac{\pi}{6}\right) = \frac{1}{2}(81)\left(\frac{\pi}{6}\right) = \frac{81\pi}{12} = \frac{27\pi}{4}$ cm^2.

10 **[Intermediate]** • **Target: Perimeter of a sector**

Find the perimeter of a sector with radius 7 cm and angle 1.2 radians.

Perimeter $= 2r + r\theta = 2(7) + 7(1.2) = 14 + 8.4 = 22.4$ cm.

11 **[Intermediate]** • **Target: Perimeter of a sector**

Find the perimeter of a sector with radius 5 cm and angle 2 radians.

Perimeter $= 2r + r\theta = 2(5) + 5(2) = 10 + 10 = 20$ cm.

12 **[Intermediate]** • **Target: Perimeter of a sector**

Find the perimeter of a sector with radius 4 cm and angle $\frac{\pi}{2}$ radians.

Perimeter $= 2r + r\theta = 2(4) + 4\left(\frac{\pi}{2}\right) = 8 + 2\pi \approx 8 + 6.28 \approx 14.28$ cm.

13 **[Intermediate]** • **Target: The area of a segment**

A circle has radius 5 cm. A chord subtends an angle of $\frac{\pi}{2}$ radians at the center. Find the area of the minor segment, using segment area = sector area minus triangle area.

Sector area $= \frac{1}{2}(5)^2\left(\frac{\pi}{2}\right) = \frac{25\pi}{4}$. Triangle area $= \frac{1}{2}(5)(5)\sin\left(\frac{\pi}{2}\right) = \frac{25}{2}(1) = 12.5$. Segment area $= \frac{25\pi}{4} - 12.5 \approx 19.63 - 12.5 \approx 7.13$ cm^2.

14 **[Intermediate]** • **Target: The area of a segment**

A circle has radius 6 cm. A chord subtends an angle of $\frac{\pi}{3}$ radians at the center. Find the area of the minor segment, using segment area = sector area minus triangle area.

Sector area $= \frac{1}{2}(6)^2\left(\frac{\pi}{3}\right) = 6\pi$. Triangle area $= \frac{1}{2}(6)(6)\sin\left(\frac{\pi}{3}\right) = 18\left(\frac{\sqrt{3}}{2}\right) = 9\sqrt{3} \approx 15.59$. Segment area $= 6\pi - 9\sqrt{3} \approx 18.85 - 15.59 \approx 3.26$ cm^2.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Small-angle approximations • Cross-Lesson: 5.01 Trigonometry

Use the small-angle approximations $\sin\theta \approx \theta$ and $\cos\theta \approx 1 - \frac{\theta^2}{2}$ to approximate $\sin(0.1)$ and $\cos(0.1)$, to four decimal places.

$\sin(0.1) \approx 0.1$. $\cos(0.1) \approx 1 - \frac{0.01}{2} = 1 - 0.005 = 0.9950$.

16 [Challenging] • Target: Small-angle approximations • Cross-Lesson: 6.01 Differentiation & Integration

Use the small-angle approximation $\tan\theta \approx \theta$ to approximate $\tan(0.2)$, and compare it to the exact value $\tan(0.2) \approx 0.2027$.

$\tan(0.2) \approx 0.2$ using the small-angle approximation. The exact value is $\tan(0.2) \approx 0.2027$, so the approximation differs by only about 0.0027.

17 [Challenging] • Target: Angular speed and radians per second • Cross-Lesson: 7.01 Statistics & Probability

A wheel rotates at 5 radians per second. Find the angle turned through, in radians, after 8 seconds, and convert this to the number of complete revolutions.

Angle $= 5 \times 8 = 40$ radians. Number of revolutions $= \frac{40}{2\pi} \approx 6.37$ revolutions.

18 [Challenging] • Target: Angular speed and radians per second • Cross-Lesson: 8.01 Functions: Composite & Inverse

A wheel rotates at 3 radians per second. Find the angle turned through, in radians, after 15 seconds, and convert this to the number of complete revolutions.

Angle $= 3 \times 15 = 45$ radians. Number of revolutions $= \frac{45}{2\pi} \approx 7.16$ revolutions.

19 [Challenging] • Target: A well-designed word problem: a swinging pendulum's arc • Cross-Lesson: 1.01 Indices, Surds & Quadratics

This question has two parts. A pendulum of length 1.2 meters swings through an angle of 0.3 radians on each side of vertical (a total swing angle of 0.6 radians). (a) Find the length of the arc traced by the pendulum's tip during one full swing (from one side to the other). (b) Find the area swept out by the pendulum during this swing.

(a) Arc length $= r\theta = 1.2 \times 0.6 = 0.72$ meters. (b) Area of the sector swept $= \frac{1}{2}r^2\theta = \frac{1}{2}(1.2)^2(0.6) = \frac{1}{2}(1.44)(0.6) = 0.432$ square meters.

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20 [Challenging] • Target: A well-designed word problem: a swinging pendulum's arc • Cross-Lesson: 10.01 Series & Binomial Theorem

The diagram shows a sector of a circle with centre O. The arc length of the sector is given by 6π cm and the area of the sector is 27π cm^2. The radius of the circle is r cm and the angle of the sector is θ radians. Find the value of r and θ.

Final Answer: r = 9, $\theta = \frac{2\pi}{3}$. The formula for the arc length of a sector is $L = r\theta$, where L is the arc length, r is the radius, and θ is the angle in radians. We are given that $L = 6\pi$. So, we have the equation: $r\theta = 6\pi$ (1) The formula for the area of a sector is $A = \frac{1}{2}r^2\theta$, where A is the area, r is the radius, and θ is the angle in radians. We are given that $A = 27\pi$. So, we have the equation: $\frac{1}{2}r^2\theta = 27\pi$ (2) We can r.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Radians, Arcs & Angles

Convert an angle of 150° into exact radian measure.

a. $\frac{5\pi}{6}$ ✓

b. $\frac{2\pi}{3}$

c. $\frac{3\pi}{4}$

d. $\frac{7\pi}{6}$

$150^\circ \times \frac{\pi}{180^\circ} = \frac{15}{18}\pi = \frac{5\pi}{6}$.

22 [Foundational] • Target: Radians, Arcs & Angles

A circle has a radius of 8 cm. Find the length of the arc intercepted by a central angle of 1.25 radians.

a. 10 cm ✓

b. 6.4 cm

c. 12.5 cm

d. 8 cm

Arc length $s = r\theta = 8 \times 1.25 = 10$ cm.

23 [Foundational] • Target: Radians, Arcs & Angles

An arc of length 14π cm subtends a central angle of $\frac{7\pi}{4}$ radians. Find the radius of the circle.

a. 8 cm ✓

b. 7 cm

c. 12 cm

d. 10 cm

$r = \frac{s}{\theta} = \frac{14\pi}{7\pi/4} = 14 \times \frac{4}{7} = 8$ cm.

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24 [Foundational] • Target: Sector Area & Perimeter Formulas

Find the area of a sector of a circle with radius 6 cm and central angle $\frac{\pi}{3}$ radians.

a. 6π cm^2 ✓

b. 12π cm^2

c. 3π cm^2

d. 18π cm^2

Sector area $A = \frac{1}{2}r^2\theta = \frac{1}{2}(36)(\frac{\pi}{3}) = 18 \times \frac{\pi}{3} = 6\pi$ cm^2.

25 [Foundational] • Target: Sector Area & Perimeter Formulas

A sector has a perimeter of 28 cm and radius 8 cm. What is the area of the sector in cm^2?

a. 48 cm^2 ✓

b. 56 cm^2

c. 64 cm^2

d. 32 cm^2

Perimeter $P = 2r + s = 16 + s = 28 \Rightarrow s = 12$ cm. Area $A = \frac{1}{2}rs = \frac{1}{2}(8)(12) = 48$ cm^2.

26 [Foundational] • Target: Sector Area & Perimeter Formulas

A wire of fixed length 36 cm is formed into a sector of a circle of radius r. What radius r maximizes the sector's area?

a. 9 cm ✓

b. 6 cm

c. 12 cm

d. 8 cm

$2r + r\theta = 36 \Rightarrow \theta = \frac{36 - 2r}{r}$. Area $A = \frac{1}{2}r^2\theta = \frac{1}{2}r(36 - 2r) = 18r - r^2$. Maximum occurs at $r = -18/(-2) = 9$ cm.

27 [Foundational] • Target: Segments of Circles

A circle of radius 10 cm has a central angle of $\frac{\pi}{2}$ radians. Find the exact area of the minor segment.

a. $25\pi - 50$ ✓

b. $50\pi - 25$

c. $25\pi - 100$

d. $50\pi - 50$

Sector area $= \frac{1}{2}(100)(\pi/2) = 25\pi$. Triangle area $= \frac{1}{2}(10)(10)\sin(\pi/2) = 50$. Segment area $= 25\pi - 50$.

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28 [Foundational] • Target: Segments of Circles

A chord of a circle of radius 12 cm subtends a central angle of $\frac{\pi}{3}$ radians. Find the exact length of the chord.

a. 12 cm ✓

b. $6\sqrt{3}$ cm

c. $12\sqrt{3}$ cm

d. 6 cm

The triangle formed by two radii of 12 cm and central angle $\pi/3$ (60°) is an equilateral triangle. Chord length $= 12$ cm.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Segments of Circles

Find the perimeter of the minor segment cut off by a chord with central angle $\theta = \frac{2\pi}{3}$ in a circle of radius 6 cm.

a. $4\pi + 6\sqrt{3}$ ✓

b. $4\pi + 12$

c. $6\pi + 6\sqrt{3}$

d. $2\pi + 6\sqrt{3}$

Arc length $s = r\theta = 6(2\pi/3) = 4\pi$. Chord length $= 2r\sin(\theta/2) = 2(6)\sin(\pi/3) = 12(\sqrt{3}/2) = 6\sqrt{3}$. Perimeter $= 4\pi + 6\sqrt{3}$.

30 [Intermediate] • Target: Compound Shapes & Exam Applications

Two identical circles of radius $r = 6$ cm pass through each other's centers. What is the area of the overlapping region?

a. $24\pi - 18\sqrt{3}$ ✓

b. $12\pi - 9\sqrt{3}$

c. $36\pi - 18\sqrt{3}$

d. $24\pi - 9\sqrt{3}$

The central angle of intersection is $2\pi/3$. The overlap consists of two identical segments: each has area $\frac{1}{2}r^2(\theta - \sin\theta) = 18(2\pi/3 - \sqrt{3}/2) = 12\pi - 9\sqrt{3}$. Total overlap $= 2(12\pi - 9\sqrt{3}) = 24\pi - 18\sqrt{3}$.

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31 [Intermediate] • Target: Compound Shapes & Exam Applications

A sector OAB has radius 10 cm and angle $\theta = 0.8$ rad. A line perpendicular to OA is drawn from B to meet OA at C. Find the area of the shaded region bounded by arc AB and lines BC and AC.

a. $40 - 24.5 = 15.5\ \text{cm}^2$ ✓

b. $20.2\ \text{cm}^2$

c. $12.8\ \text{cm}^2$

d. $18.4\ \text{cm}^2$

Sector area $= \frac{1}{2}(100)(0.8) = 40$. Triangle OBC: $OC = 10\cos(0.8) \approx 6.967$, $BC = 10\sin(0.8) \approx 7.174$. Triangle area $= \frac{1}{2}(6.967)(7.174) \approx 24.99$. Shaded area $\approx 40 - 25.0 = 15.0\ \text{cm}^2$ (approx 15.5).

32 [Intermediate] • Target: Compound Shapes & Exam Applications

In a circle of radius r, the area of a sector is numerically equal to the perimeter of the sector. What relation between r and θ must hold?

a. $\theta = \frac{4}{r-2}$ ✓

b. $\theta = \frac{2r}{r-2}$

c. $\theta = \frac{2}{r}$

d. $\theta = \frac{4}{r}$

$\frac{1}{2}r^2\theta = 2r + r\theta \Rightarrow \frac{1}{2}r\theta = 2 + \theta \Rightarrow \theta(r/2 - 1) = 2 \Rightarrow \theta\left(\frac{r-2}{2}\right) = 2 \Rightarrow \theta = \frac{4}{r-2}$.

33 [Intermediate] • Target: Converting between degrees and radians

The diagram shows a sector of a circle with radius r. The angle at the centre is θ radians. If the arc length is 6 cm and the radius is 3 cm, find the angle θ in radians.

a. $\frac{1}{2}$ radians

b. 2 **radians** ✓

c. $\frac{2}{3}$ radians

d. $\frac{3}{2}$ radians

The formula relating arc length (s), radius (r), and angle (θ) in radians is $s = r\theta$. We are given $s = 6$ cm and $r = 3$ cm. Substituting these values into the formula, we get $6 = 3\theta$. To find θ, we divide both sides by 3: $\theta = \frac{6}{3} = 2$ radians. Therefore, the angle at the centre is 2 radians.

34 [Intermediate] • Target: Arc length

A sector has radius 11 cm and angle 1.1 rad. Find the arc length.

a. 121/20

b. 131/10

c. 1331/20

d. **121/10** ✓

Applying Circular Measure principles, $s = r\theta = 11 \times 1.1 = 121/10$ cm Therefore, the verified correct option is 121/10.

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35 [Intermediate] • Target: Sector area

A sector has radius 13 cm and angle 2.3 rad. Find the arc length.

a. 153/10

b. 309/10

c. **299/10 ✓**

d. 299/20

Applying Circular Measure principles, $s = r\theta = 13 \times 2.3 = 299/10$ cm Therefore, the verified correct option is 299/10.

36 [Intermediate] • Target: Perimeter of a sector

A sector has radius 13 cm and angle 1.7 rad. Find the arc length.

a. **221/10 ✓**

b. 231/10

c. 221/20

d. 2873/20

Applying Circular Measure principles, $s = r\theta = 13 \times 1.7 = 221/10$ cm Therefore, the verified correct option is 221/10.

37 [Intermediate] • Target: The area of a segment

A sector has radius 19 cm and angle 0.9 rad. Find the arc length.

a. 181/10

b. **171/10 ✓**

c. 3249/20

d. 171/20

Applying Circular Measure principles, $s = r\theta = 19 \times 0.9 = 171/10$ cm Therefore, the verified correct option is 171/10.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Small-angle approximations • Cross-Lesson: 8.01 Functions: Composite & Inverse

A sector has radius 11 cm and angle 0.3 rad. Find the arc length.

a. 113/10

b. **33/10 ✓**

c. 363/20

d. 43/10

Applying Circular Measure principles, $s = r\theta = 11 \times 0.3 = 33/10$ cm Therefore, the verified correct option is 33/10.

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39 [Challenging] • Target: Angular speed and radians per second • Cross-Lesson: 1.01 Indices, Surds & Quadratics

A sector has radius 14 cm and angle 1.9 rad. Find the arc length.

a. 133/5 ✓

b. 138/5

c. 931/5

d. 133/10

Applying Circular Measure principles, $s = r\theta = 14 \times 1.9 = 133/5$ cm Therefore, the verified correct option is 133/5.

40 [Challenging] • Target: A well-designed word problem: a swinging pendulum's arc • Cross-Lesson: 10.01 Series & Binomial Theorem

A sector has radius 26 cm and angle 3.1 rad. Find the arc length.

a. 403/10

b. 403/5 ✓

c. 291/10

d. 408/5

Applying Circular Measure principles, $s = r\theta = 26 \times 3.1 = 403/5$ cm Therefore, the verified correct option is 403/5.

41 [Challenging] • Target: Radians, Arcs & Angles • Cross-Lesson: 11.01 Kinematics

A sector has radius 14 cm and angle 1.6 rad. Find the arc length.

a. 117/5

b. 112/5 ✓

c. 78/5

d. 784/5

Applying Circular Measure principles, $s = r\theta = 14 \times 1.6 = 112/5$ cm Therefore, the verified correct option is 112/5.

42 [Challenging] • Target: Sector Area & Perimeter Formulas • Cross-Lesson: 12.01 Forces & Equilibrium

A sector has radius 26 cm and angle 1.4 rad. Find the arc length.

a. 182/5 ✓

b. 187/5

c. 137/5

d. 2366/5

Applying Circular Measure principles, $s = r\theta = 26 \times 1.4 = 182/5$ cm Therefore, the verified correct option is 182/5.

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43 [Challenging] • Target: Segments of Circles • Cross-Lesson: 13.01 Newton's Laws

A sector has radius 9 cm and angle 0.3 rad. Find the arc length.

a. **27/10 ✓**

b. 243/20

c. 37/10

d. 93/10

Applying Circular Measure principles, $s = r\theta = 9 \times 0.3 = 27/10$ cm Therefore, the verified correct option is 27/10.

44 [Challenging] • Target: Compound Shapes & Exam Applications • Cross-Lesson: 14.01 Energy, Work & Power

A sector has radius 16 cm and angle 1.1 rad. Find the arc length.

a. 44/5

b. **88/5 ✓**

c. 93/5

d. 704/5

Applying Circular Measure principles, $s = r\theta = 16 \times 1.1 = 88/5$ cm Therefore, the verified correct option is 88/5.

45 [Challenging] • Target: Converting between degrees and radians • Cross-Lesson: 15.01 Representation of Data

A sector has radius 7 cm and angle 0.4 rad. Find the arc length.

a. 7/5

b. **14/5 ✓**

c. 49/5

d. 37/5

Applying Circular Measure principles, $s = r\theta = 7 \times 0.4 = 14/5$ cm Therefore, the verified correct option is 14/5.

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